

lgbm_3class_xfn5d_35f — Model Card

LightGBM

3-Class

35 Features

TD Bank directional signal detector — predicts 5-day excess return vs XFN sector ETF · Locked 2026-04-22

60.5%

Active sign accuracy

0.572

Dir AUC
(random=0.50)

96.8%

Active coverage

35

Features (30+5 TA)

7

Validation folds

Problem Framing

Given information available on day d , the model outputs one of three signals for the next 5 trading days. The target is TD's 5-day return minus XFN's 5-day return — removing the positive-drift bias that makes raw-return models trivially beatable by always-up rules.

Signal	Label	Condition	Interpretation
Outperform	+1	Excess return > +0.30%	Take long position vs sector
Neutral	0	Excess return within ±0.30%	No actionable signal
Underperform	−1	Excess return < −0.30%	Reduce / hedge vs sector

Model Architecture

4

num_leaves (shallow)

80

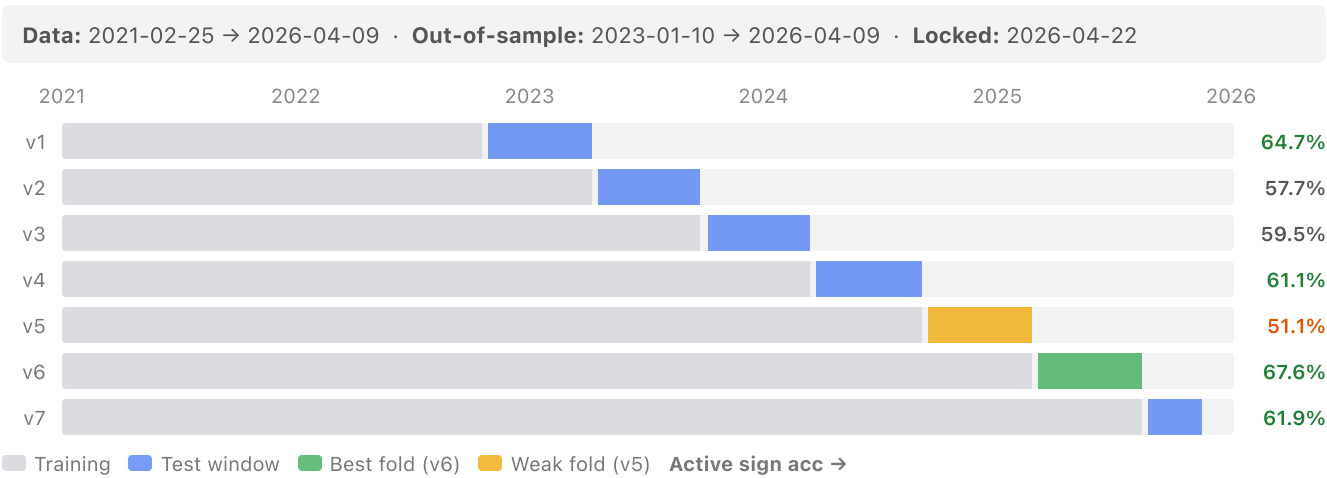
n_estimators

40

min_child_samples

Hyperparameters selected via 36-config grid search ($\text{num_leaves} \in \{4,8,16,31\}$, $\text{min_child_samples} \in \{20,40,60\}$, $\text{n_estimators} \in \{80,120,200\}$) across all 7 folds. Composite score = $0.6 \times \text{ASA} + 0.4 \times \text{Macro F1}$. Shallower trees ($\text{num_leaves}=4$) proved optimal when features expanded from 30 to 35.

Training & Holdout Timeline



Per-Fold Performance

Fold	Train window	Test window	Days	Mean acc	Macro F1	Active sign acc	Dir AUC	Coverage
v1 (H1 2023)	2021-02-25 → 2022-12-30	2023-01-10 → 2023-06-30	121	50.4%	0.419	64.7%	0.634	90.9%
v2 (H2 2023)	2021-02-25 → 2023-06-30	2023-07-11 → 2023-12-29	119	47.0%	0.363	57.7%	0.599	95.8%
v3 (H1 2024)	2021-02-25 → 2023-12-29	2024-01-09 → 2024-06-28	121	47.9%	0.243	59.5%	0.469	100.0%
v4 (H2 2024 — AML peak)	2021-02-25 → 2024-06-28	2024-07-09 → 2024-12-31	121	52.9%	0.380	61.1%	0.640	100.0%
v5 (H1 2025 — weakest)	2021-02-25 → 2024-12-31	2025-01-09 → 2025-06-30	120	40.0%	0.296	51.1%	0.532	90.8%
v6 (H2 2025 — best)	2021-02-25 → 2025-06-30	2025-07-09 → 2025-12-31	121	52.8%	0.392	67.6%	0.626	100.0%
v7 (Q1 2026)	2021-02-25 → 2025-12-31	2026-01-09 → 2026-04-09	63	46.3%	0.335	61.9%	0.503	100.0%
7-fold mean				48.2%	0.347	60.5%	0.572	96.8%

7-Fold Average vs Baselines

lgbm_3class_xfn5d_35f
60.5%
Active sign acc
0.572
Dir AUC
96.8%
Coverage
48.2%
Mean accuracy
0.347
Macro F1

Majority Baseline
38.5%
Active sign acc
100%
Coverage
38.5%
Mean accuracy
Always predict training majority class

Momentum Baseline
39.0%
Active sign acc
79%
Coverage
36.6%
Mean accuracy
Predict continuation of TD vs XFN 5d return

Always-Long Reference
43%
Active sign acc
100%
Coverage
43%
Mean accuracy
Not viable for pension — no drawdown protection

+22pp lift over majority and momentum baselines on active sign accuracy
lgbm_3class_xfn5d_35f is directionally correct 60.5% of the time it takes a position — vs 38–39% for both pension-relevant baselines. Adding 5 TA features improved ASA by +0.5pp and Dir AUC by +0.6pp vs the 30-feature baseline, while coverage increased to 96.8%.

Feature Set (35)

NLP event features computed from LLM-scored earnings call transcripts, forward-filled and exponentially decayed by `days_since_call` (half-life 20 days). TA/Qlib additions selected via diagnostic SHAP ranking across 25 candidates.

Market / Price (8)
`td_return_5d` `td_return_20d` `td_volatility_20d`
`td_vs_xfn_5d` `td_vs_tsx_5d` `td_corr_pv_20d`
`td_volume_change_20d` `td_dist_52w_high`

NLP Event — decayed (11)
`evt_ceo_tone` `evt_cfo_tone` `evt_framing_gap`
`evt_aml_pressure` `evt_aml_shift`
`evt_guidance_strength` `evt_guidance_shift`
`evt_macro_topic` `evt_topic_entropy`
`evt_news_tone` `evt_news_flow`

Rates / Macro (2) + Global Risk (4) + News Flow (3) + Timing (2)

- yield_curve_slope
- yield_10y_level
- vix_volatility_20d
- dxy_level
- gold_level
- fx_usdcad_level
- news_sent_mean_30d
- news_count_30d
- days_since_last_news
- days_since_call
- is_earnings_week

TA / Qlib Additions (5) — new in 35f

- td_rsqr_60d
- td_wvma_5d
- td_wvma_20d
- td_corr_pv_5d
- td_beta_20d

Selected via fold-local SHAP ranking across 25 TA candidates

Limitations

#	Limitation	Detail
1	Single stock	Trained only on TD Bank — signals are specific to TD's dynamics and earnings cadence
2	Directional only	Does not predict magnitude, only direction relative to XFN sector
3	AML feature timing	evt_aml_pressure/shift were included with domain knowledge; documented in experiment log
4	Data dependency	Requires daily feature pipeline (model_features_daily.parquet) and price data
5	TA feature selection	5 TA additions selected via SHAP ranking; modest improvement (+0.5pp ASA) on ~1,285 rows

Artifact: artifacts/lgbm_3class_xfn5d_35f_tuned-2026-04-22.pkl · Generated 2026-04-22